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DEEPFAKE DETECTION

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Deepfake Detection

Abstract

A deepfake is a synthetic image, video, or audio clip generated by deep neural networks to resemble a real person convincingly. Advances in autoencoder- and Generative Adversarial Network (GAN)-based architectures have made such media cheap to produce and difficult to distinguish from authentic recordings, creating a growing risk to journalism, elections, financial systems, and personal privacy. This report examines the deepfake problem from generation through detection to prevention. We first describe the encoder-decoder pipeline that produces synthetic media and the objective functions that drive it. We then present two complementary detection strategies: passive detection, which inspects spatial, temporal, and audio-visual artifacts left behind by the generation process, and proactive defense, which embeds cryptographic watermarks and provenance metadata at the point of creation so that origin can be verified later. We compare the two strategies, outline where each is used in practice, and argue that no single detector is a durable solution because detection and generation are locked in an adversarial cycle. We close by discussing the shift toward content provenance standards such as C2PA and the open challenges that remain.

1 Introduction

A **deepfake** is a high-quality artificial media: either an image, video, or sound, in which artificial neural networks have been used to modify, replace, or create a digital image of a person. The term "deep learning" and "fake" is combined to indicate that such media now can be created with less effort than before: previously, it took much time and experience to make a convincing digital fake, while now, algorithms, trained on a set of digital media, can learn the patterns of a particular person's image or sound and generate new material impersonating this person.

While generative AI could be used for good, such as dubbing foreign films or storing information about history, its availability for everyone has compromised confidence in the global information ecosystem. The ability to create forgeries has fundamentally undermined trust in media as a whole, resulting in a documented crisis of trust. Highlighting the scale of this issue, the *Deepfakes Statistics — 2026 Sourced Report* reveals that 74% of global consumers are now worried about the societal impact of deepfakes, with synthetic media projected to account for up to 70% of crypto-related crimes in the near future [3]. This could take the form of false geopolitical information that has been made to influence the masses, or even fraud, such as using someone's voice to withdraw funds. There are many other ways that confidence in the global information ecosystem could be compromised. On a personal level, people may fear that their digital identities are

being misused or abused. Moreover, there are reports of a “Liar’s Dividend” – the possibility that forgeries have created an environment of distrust so that genuine evidence of someone’s wrongdoing might itself be dismissed as fake.

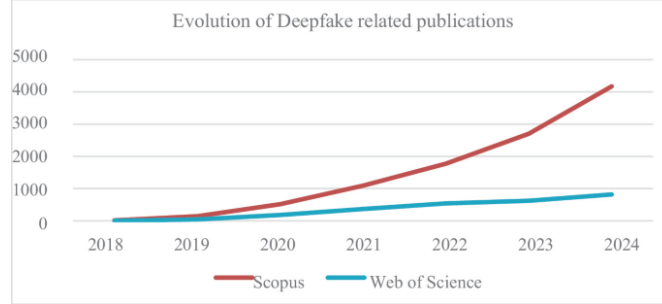


Figure 1: Evolution of deepfake-related publications from 2018 to 2024 across Scopus and Web of Science databases, illustrating the rapid acceleration of academic focus on synthetic media.

This report examines the deepfake ecosystem by addressing three core stages of the synthetic media life cycle:

- **Generation:** How is synthetic media architected and mathematically optimized? (Section 2)
- **Detection:** How can algorithms identify the spatial and temporal artifacts left behind by generative models? (Section 3)
- **Prevention:** How can cryptographic provenance and watermarking verify media origin before deception occurs? (Section 4)

We conclude by evaluating the adversarial limitations of passive detection, surveying real-world deployment challenges, and exploring the necessity of a layered defense-in-depth strategy (Sections 5–7).

2 Generation of Deepfake Media

In order to detect synthetic media, it is essential to understand what processes are used to create them. This section will describe standard deep learning architectures used for identity swapping and reenactment, as well as the mathematical objectives which they are optimized towards.

2.1 General Architecture

The majority of contemporary face-swapping and voice-cloning systems rely on an encoder-decoder framework, frequently augmented by adversarial training. The pipeline, illustrated in Figure 2, requires two distinct inputs: a *source identity* (the target face or voice to be injected) and a *target context* (the background, lighting, and semantic motion of the destination media).

The deep learning engine is tasked with decoupling these attributes. The encoder maps the inputs into a lower-dimensional *latent representation*, effectively isolating the source’s biometric identity from the target’s expressions and environmental context. The decoder

then reconstructs this compressed data into a synthetic output that flawlessly superimposes the source identity onto the target’s movements and environment.

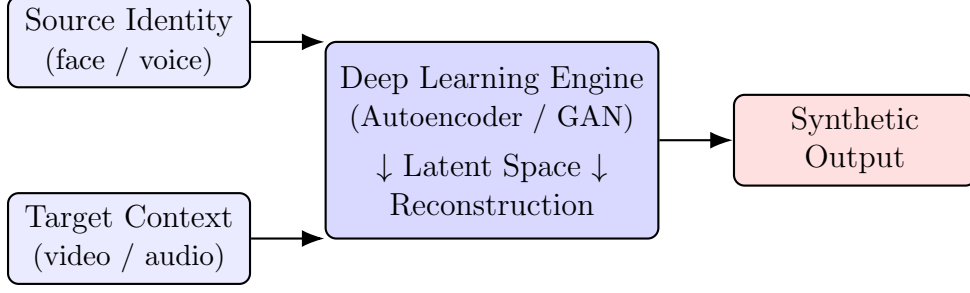


Figure 2: General deepfake generation pipeline. A source identity and a target context are compressed into a shared latent space and decoded into a unified synthetic output.

2.2 Mathematical Formulation

Deepfake generation relies on deep neural networks to learn representations of facial appearance, identity, and motion from training data. Among the earlier approaches to face swapping, autoencoder-based architectures played an important role, while more recent systems have incorporated adversarial learning and other generative techniques [18, 14, 13].

Let x denote an input sample, such as a facial image or an aligned video frame, drawn from a training dataset $\mathcal{D}$. An autoencoder consists of an encoder E_ϕ and a decoder G_θ , parameterized by ϕ and θ , respectively. The encoder maps the input into a lower-dimensional latent representation,

$$z = E_\phi(x), \quad (1)$$

while the decoder reconstructs the input from this representation,

$$\hat{x} = G_\theta(z). \quad (2)$$

The network can be trained by minimizing a reconstruction loss that measures the difference between the original and reconstructed samples. A commonly used formulation is the mean squared error:

$$\mathcal{L} * \text{recon} = \mathbb{E} * x \sim \mathcal{D} [\|x - \hat{x}\|_2^2]. \quad (3)$$

This objective encourages the encoder–decoder pair to preserve the important visual information required to reconstruct the input [9].

For face-swapping applications, a well-known DeepFakes architecture uses a shared encoder together with separate decoders for the source and target identities. The shared encoder learns a representation that can be used for both identities, while the individual decoders learn to reconstruct the corresponding facial appearance. During inference, the latent representation of one person’s face can be passed through the decoder associated with another identity, producing a synthesized face with the target identity while retaining aspects of the source facial configuration [16, 14]. This encoder–decoder formulation is one of the classical foundations of modern face-swapping systems.

A second important family of generative models is the Generative Adversarial Network (GAN), introduced by Goodfellow et al. [7]. A GAN consists of a generator G_θ and a

discriminator D_ψ . The generator attempts to produce synthetic samples that resemble real data, whereas the discriminator attempts to distinguish real samples from generated ones. The original GAN formulation is expressed as the following minimax objective:

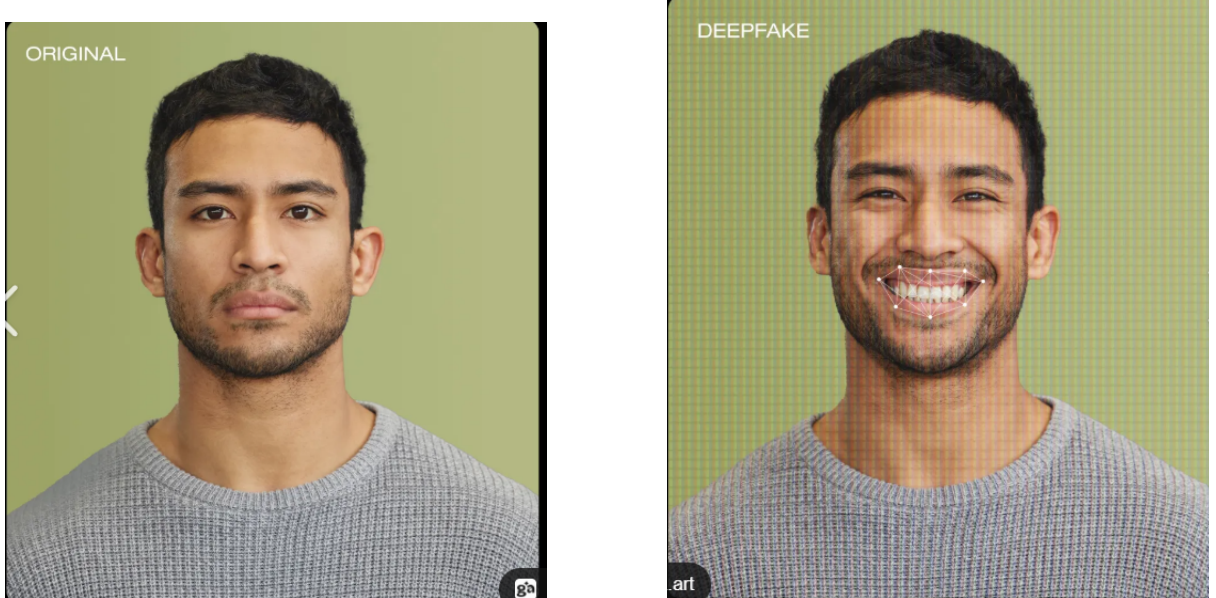
$$\min_{\theta} \max_{\psi} V(D_\psi, G_\theta) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D_\psi(x)] + \mathbb{E}_{z \sim p(z)} [\log (1 - D_\psi(G_\theta(z)))]. \quad (4)$$

Here, $p_{\text{data}}(x)$ represents the distribution of real training samples and $p(z)$ denotes the prior distribution from which the generator’s latent input is sampled. The discriminator maximizes this objective by correctly distinguishing real and generated samples, while the generator minimizes it by producing samples that are increasingly difficult for the discriminator to distinguish from real data [7]. Adversarial training can therefore encourage greater perceptual realism in generated facial content, although GANs represent only one of several generative approaches used in deepfake systems [13].

2.3 Visual Comparison: Original vs. Deepfake

To illustrate the output of the generative pipelines discussed above, Figure 3 provides a side-by-side visual comparison of an authentic facial image and its synthetically manipulated counterpart. While the original image captures natural biometric features, realistic skin textures, and physically accurate environmental lighting, the synthetic deepfake alters these attributes to superimpose a new identity.

Although modern adversarial networks are capable of synthesizing highly convincing forgeries that easily deceive the human eye, forensic analysis often reveals microscopic inconsistencies in the manipulated image. Subtle artifacts—such as unnaturally smoothed skin, inconsistent reflections, or blending errors around the facial boundaries—serve as the foundational signals for the detection algorithms discussed in Section 3.



(a) Original Image

(b) Synthetic Deepfake

Figure 3: A side-by-side comparison of an authentic frame and a manipulated deepfake, demonstrating both the photorealism of generative models and the subtle artifacts they leave behind.

3 Passive Detection: Finding the Flaws

Detection strategies fall into two broad categories, summarized in Table 1: *passive detection*, which examines media after it has been created, and *proactive defense*, which embeds safeguards into the generation process itself. This section covers passive detection; proactive defense is discussed in Section 4.

Aspect	Passive Detection	Proactive Defense
When it acts	After media has been created	At the moment media is created
Method	Examines media for hidden artifacts left by the generation process	Embeds a cryptographic watermark or provenance record into the media
Posture	Reactive	Preventative
Typical question	<i>“Does this look fake?”</i>	<i>“Can we prove where this came from?”</i>

Table 1: Comparison of passive detection and proactive defense.

3.1 Spatial Artifact Detection

A single frame is often enough to reveal a deepfake. Generative models tend to leave two characteristic spatial artifacts. First, skin regions are frequently over-smoothed because the decoder averages over many training examples, producing a waxy texture that lacks natural pore-level detail. Second, generated faces often exhibit an unnaturally even focus: the face and the background are equally sharp, whereas a real photograph taken with a camera lens shows a shallow depth of field that blurs the background relative to the subject.

These cues can be learned automatically. A Convolutional Neural Network (CNN) is trained as a binary classifier $f_w : x \mapsto p \in [0, 1]$, where x is an input frame, w denotes the network’s weights, and p is the predicted probability that the frame is synthetic. Given a labeled training set of real and fake frames with ground-truth labels $y \in \{0, 1\}$, the network is trained by minimizing the binary cross-entropy loss

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)], \quad (5)$$

where N is the number of training examples in a mini-batch. Figure 4 illustrates the two artifacts described above together with the classification pipeline.

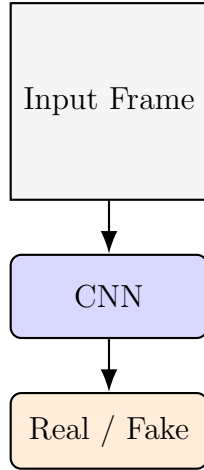


Figure 4: A CNN classifier trained to distinguish real frames from synthetic ones using spatial cues such as skin smoothness and depth-of-field consistency.

3.2 Temporal Artifact Detection

A single frame can look convincing while the video as a whole does not, because generative models operate mostly on a frame-by-frame basis and are not always consistent across time. Three temporal cues are commonly exploited.

- **Blink rate:** humans blink at a fairly regular interval; early face-swap datasets under-represented closed-eye frames, causing generated subjects to blink too rarely or too regularly.
- **Motion smoothness:** synthetic faces can exhibit subtle jitter or warping between consecutive frames that a real camera recording would not produce.
- **Physiological signal:** a real face shows faint, periodic color shifts caused by blood flow beneath the skin (remote photoplethysmography); this signal is difficult for a generator to reproduce consistently.

Because these cues depend on the relationship between frames rather than on any single frame, they are best captured by a sequence model such as a Long Short-Term Memory (LSTM) network or a Vision Transformer applied across a short clip $\{x_1, x_2, \dots, x_T\}$, where T is the number of frames in the clip. The sequence model is trained with the same binary cross-entropy objective as Equation 5, but the input is now a temporal window rather than a single frame.

3.3 Multimodal Detection

Voice can be cloned just as convincingly as a face can be swapped, and an attacker who fakes only one modality often leaves the other unchanged, which creates an inconsistency that can be measured directly. The most reliable cue in this category is lip-sync mismatch: the shape of the mouth in the video should match the phoneme being produced in the audio at every instant, and a mismatch of even a few frames is detectable even when neither the video nor the audio is individually flagged as fake by a single-modality detector.

A multimodal detector processes the video stream and the audio stream separately, using a video encoder for the former and an audio encoder in the style of Whisper or wav2vec2 for the latter, and then combines the two streams with a cross-modal transformer that checks whether the acoustic and visual signals are consistent with each other. This cross-checking approach is powerful precisely because it does not require either modality to be individually recognizable as synthetic; it only requires the two to disagree.

3.4 The Cat-and-Mouse Problem

Passive detection faces a structural limitation. Every new detection technique that becomes public gives generator designers a specific artifact to correct in the next model version, so detection accuracy that is measured on today’s fakes routinely degrades on tomorrow’s. This dynamic mirrors the adversarial training objective of Equation 4 at the scale of the research community: detectors act as an external discriminator, and generators are iteratively improved to defeat them. Because detection is always one step behind the latest generation technique, passive detection alone cannot be a permanent solution; it must be paired with a proactive strategy, which we discuss next.

4 Proactive Defense: Proving Provenance

Rather than trying to spot a fake after it exists, a proactive defense makes authenticity verifiable from the moment media is created.

4.1 Invisible Watermarking

An invisible watermark is a pattern embedded directly into a generated image or video at creation time, using a cryptographic key k known to the issuing organization. The watermark embedding function can be described as $x_w = W_k(x)$, where x is the generator’s raw output, x_w is the watermarked output, and W_k is the embedding operator. The pattern is designed to be perceptually invisible, so x_w and x look identical to a human viewer, while remaining statistically detectable to a verification system that has access to the key. A well-designed watermark is also robust: it should survive common transformations such as compression, cropping, and re-encoding through a screenshot, so that the provenance signal is not lost as content is shared across platforms.

Verification is the mirror operation of embedding: given a piece of media and the same key k , a verification system extracts a candidate pattern and checks it for a match. Figure 5 summarizes the pipeline from generation through verification.

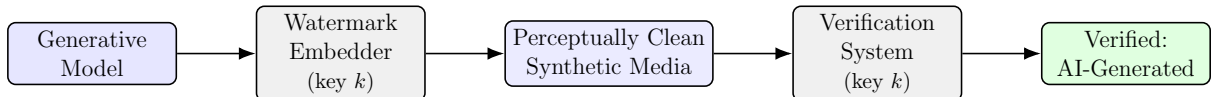


Figure 5: Invisible watermarking pipeline: a key-dependent pattern is embedded at generation time and later extracted by a verification system.

Watermarking in Action

As a concrete industrial deployment of generation-time watermarking, Anthropic’s Claude models offer an illustrative example. Anthropic’s implementation is a version of SynthID-

Text, the method Google DeepMind published in Nature in 2024 and deployed on Gemini, demonstrating the convergence of independent AI developers on a shared watermarking paradigm [4].

The mechanism operates at the token-sampling stage rather than as a post-hoc transformation of the output. During autoregressive generation, a language model frequently faces multiple plausible continuations at a given position (e.g., completing "The sky was" with either "grey" or "cloudy"). In an unwatermarked system, ties among such near-equivalent candidates are resolved via an ordinary pseudorandom number generator. Anthropic's watermarking system instead uses a secret cryptographic key combined with information from the preceding tokens to influence this selection, biasing an otherwise-arbitrary choice according to a hidden, deterministic function of the key and local context. Critically, this approach preserves output fidelity: every word in a watermarked output remains a word the model could have produced anyway, with nothing added to the text and no hidden characters inserted. Detectability is therefore statistical rather than lexical — no individual sentence appears unusual, and the signal exists only in aggregate across a sufficiently long span of text, requiring roughly 150–200 words for reliable detection. A party holding the key can, in principle, measure agreement between the observed token choices and those the keyed process would have generated, yielding a probabilistic estimate of authorship, while the pattern remains statistically invisible to any party lacking the key.

This case illustrates a central trade-off in text watermarking research: robustness to copy-paste propagation and imperceptibility to human readers, at the cost of detection reliability on short passages.

4.2 Content Provenance Standards

Watermarking answers a narrow question: was this specific pattern present? A broader approach, known as content provenance, attaches a signed record of an asset's history, its origin, its creator, and every edit it has undergone, directly to the file as metadata. The Coalition for Content Provenance and Authenticity (C2PA) is the leading industry standard in this space; it defines *Content Credentials* that travel with a media file and can be inspected by any compliant viewer. Table 2 contrasts this provenance-based approach with the detection-based approach of Section 3.

Aspect	Detection	Provenance
Question asked	Does this look fake?	Can we prove where this came from?
Basis	Statistical inference over artifacts	Cryptographic signature and metadata
Confidence	Probabilistic; can be wrong	Deterministic, if the signature is intact
Weakness	Loses ground as generators improve	Requires adoption across creation tools and platforms

Table 2: Detection asks whether content looks fake; provenance asks whether its origin can be proven.

5 Applications and Defense in Depth

5.1 Where Detection Matters

The guarantees required of a detection system differ sharply by domain, as summarized in Table 3.

Domain	Typical Use	Primary Requirement
Social platforms	Upload scanning, content labeling	High throughput, low latency
FinTech and identity	Fraud prevention, liveness checks	Low false-negative rate
Courts and forensics	Evidence auditing, forensic verification	High explainability, legal defensibility
Government and elections	Statement verification, media integrity	Public trust, rapid response

Table 3: Representative application domains for deepfake detection and their primary requirements.

5.2 Defense in Depth

No single technique from Sections 3 and 4 is sufficient on its own. In practice, an effective system layers three stages, illustrated in Figure 6: *prevention*, which limits access to generation tools and embeds watermarks; *detection*, which screens content already in circulation using spatial, temporal, and audio-visual cues; and *verification*, which checks cryptographic provenance when it is available. Each stage compensates for the weaknesses of the others, so that a failure at one stage does not compromise the whole system.



Figure 6: Defense in depth: prevention, detection, and verification act as three complementary layers rather than a single point of failure.

Algorithm 1 summarizes the operational logic of a layered detection service that combines a passive classifier with provenance verification.

Algorithm 1 Layered Screening of an Uploaded Media Item

Require: Media item x , watermark key k , spatial detector f_w , temporal detector g_v , decision threshold τ

- 1: Attempt to extract a provenance record or watermark from x using key k
- 2: **if** a valid, matching record is found **then**
- 3: **return** VERIFIED AUTHENTIC
- 4: **else**
- 5: Compute spatial score $p_s \leftarrow f_w(x)$ (Equation 5)
- 6: Compute temporal score $p_t \leftarrow g_v(x)$ over a short clip
- 7: Combine scores: $p \leftarrow \frac{1}{2}(p_s + p_t)$
- 8: **if** $p > \tau$ **then**
- 9: **return** FLAGGED AS LIKELY SYNTHETIC
- 10: **else**
- 11: **return** PASSED, NO PROVENANCE RECORD
- 12: **end if**
- 13: **end if**

6 Challenges and Future Directions

Sections 3 and 4 show what passive detection and proactive provenance can do under controlled conditions; three difficulties determine whether they hold up in deployment. First, detectors generalize poorly across generators: a systematic review of cross-dataset benchmarks found accuracy declines of roughly 11% for transformer-based detectors and over 15% for CNN-based detectors when tested against an unseen generator family [15], and the field’s response is domain-generalized and meta-learning detectors trained to be invariant to generator identity rather than tuned to one [12]. Second, a detector or a watermark can be evaded deliberately rather than merely outdated: small, imperceptible perturbations have been shown to flip a detector’s decision or strip an embedded watermark, and such attacks transfer across schemes the adversary never directly queried [10]. Third, the trust infrastructure behind provenance is itself imperfect: an independent 2026 security analysis found the C2PA specification incomplete and open to forgery in places, with one hardware vendor forced to revoke every content credential it had issued after a signing flaw [6] – even as the European Union’s AI Act now legally requires this kind of provenance marking, with Article 50 obligations in force from 2 August 2026 and fines of up to 3% of global turnover [5]. Table 4 summarizes these challenges alongside the field’s response.

Challenge	Evidence	Response
Poor cross-generator generalization	Cross-dataset AUC drops $\sim 11\%$ (transformer) to $>15\%$ (CNN) [15]	Domain-generalized, meta-learning detectors [12]
Adversarial evasion of detectors/watermarks	Imperceptible perturbations flip decisions or strip watermarks [10]	Adversarially hardened detectors and watermarks
Gaps in provenance trust infrastructure	C2PA found incomplete; credential revocations have occurred [6]	Independent audits, stronger CA infrastructure
Regulation outpacing technical readiness	EU AI Act Article 50 mandates marking from Aug. 2026 [5]	Standardized provenance with detection as fallback
Risk of full automation	Legal/ethical risk without human review (Section 5.1)	Human-in-the-loop review

Table 4: Key challenges in deepfake detection and the corresponding direction of response.

These directions converge on the same picture: detection that is multimodal and generator-agnostic rather than tuned to one architecture, provenance that is independently audited rather than merely deployed, and human judgment retained for consequential decisions.

7 Conclusion

Deepfake generation and detection are two sides of the same adversarial process described by Equation 4: as generators improve, detectors must improve to match them, and the cycle repeats. Passive detection, whether based on spatial artifacts (Section 3.1), temporal inconsistency (Section 3.2), or cross-modal mismatch (Section 3.3), remains useful but cannot win this race alone: it always responds to a technique that already exists, and, as Section 6 showed, its accuracy transfers only partially from one generator to the next. Proactive defense through watermarking and content provenance (Section 4) reframes the question from “does this look fake?” to “can we prove where this came from?”, which is what makes verification durable as generation techniques advance – though neither a detector nor a watermark is beyond deliberate evasion, and even the provenance standards meant to anchor this side still carry documented security gaps.

The goal, then, is not a perfect detector; no fixed classifier stays accurate indefinitely against a generator, or an adversary, built specifically to defeat it. The realistic goal is a layered ecosystem combining prevention, detection, and verification (Section 5.2), each independently audited and continuously updated, so that fabricated media is harder to spread convincingly and genuine media is easier to verify. With the EU AI Act now requiring machine-readable marking of synthetic content [5], this shift is no longer only academic: provenance infrastructure that was optional at the start of this report is fast becoming a legal expectation. As the technology and its governance mature together, we expect the balance to keep shifting from after-the-fact detective work toward built-in, verifiable authenticity.

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